

C.2.3 Modifications and Chemical Features

Field Name	Full Name	Description	Quant.	LLM
ft_carbohyd	Carbohydrate Modifications	Locations where sugar groups are attached to the protein	Y	Y
ft_lipid	Lipid Modifications	Sites where lipid molecules are attached to the protein	Y	Y
ft_mod_res	Modified Residues	Amino acids that undergo post-translational modifications	Y	Y
cc_cofactor	Cofactor Information	Non-protein molecules required for protein function	N	Y

C.2.4 Targeting and Localization

Field Name	Full Name	Description	Quant.	LLM
ft_signal	Signal Peptide	Sequence that directs protein trafficking in the cell	Y	Y
ft_transit	Transit Peptide	Sequence guiding proteins to specific cellular compartments	Y	Y

C.2.5 Functional Domains and Regions

Field Name	Full Name	Description	Quant.	LLM
ft_compbias	Compositionally Biased Regions	Sequences with unusual amino acid distributions	Y	Y
ft_domain	Protein Domains	Distinct functional or structural protein units	Y	Y
ft_motif	Short Motifs	Small functionally important amino acid patterns	Y	Y
ft_region	Regions of Interest	Areas with specific biological significance	Y	Y
ft_zn_fing	Zinc Finger Regions	DNA-binding structural motifs containing zinc	Y	Y
ft_dna_bind	DNA Binding Regions	Regions that interact with DNA	N	Y
ft_repeat	Repeated Regions	Repeated sequence motifs within the protein	N	Y
cc_domain	Domain Commentary	General information about functional protein units	N	Y

C.2.6 Functional Annotations

Field Name	Full Name	Description	Quant.	LLM
cc_catalytic_activity	Catalytic Activity	Description of the chemical reaction(s) performed by the protein	N	Y
ec	Enzyme Commission Number	Enzyme Commission number for categorizing enzyme-catalyzed reactions	N	Y
cc_activity_regulation	Activity Regulation	Information about how the protein's activity is controlled	N	Y
cc_function	Function	General description of the protein's biological role	N	Y
protein_families	Protein Families	Classification of the protein into functional/evolutionary groups	N	Y
go_f	Gene Ontology Function	Gene Ontology terms describing molecular functions	N	Y

Active Site	Compositional Bias	Signal Peptide
• Acyl-ester intermediate • O-(3'-phospho-DNA)-tyrosine intermediate • Tele-phosphohistidine intermediate	• Acidic residues • Pro residues	• Tat-type signal • any
Coiled Coil	Disulfide Bond	Transit Peptide
	Modified Residue	• Mitochondrion • any
	• 4-aspartylphosphate • O-(pantetheine 4'-phosphoryl)serine	Zinc Finger
		• CR-type • PHD-type • RING-type • any
	Domain [FT]	
• 2Fe-2S ferredoxin-type • 4Fe-4S ferredoxin-type 1 • 4Fe-4S ferredoxin-type 2 • AB hydrolase-1 • ABC transmembrane type-1* • ABC transporter* • ATP-grasp • C-type lectin • C2 • CBS 1 • CBS 2 • CN hydrolase • CP-type G • Carrier • CheB-type methylesterase • Core-binding (CB) • DPKK • DRBM • Disintegrin • EngA-type G 2 • EngB-type G • Era-type G • Exonuclease • F-box* • FAD-binding FR-type • FAD-binding PCMH-type • Fe2OG dioxygenase • Fibronectin type-III 1 • Fibronectin type-III 2 • G-alpha • GH16 • GH18 • GMPS ATP-PPase	• GST C-terminal • GST N-terminal • Glutamine amidotransferase type-1* • HD • HTH araC/xylS-type* • HTH cro/C1-type • HTH luxR-type • HTH lysR-type • HTH marR-type • HTH tetR-type • Helicase ATP-binding • Helicase C-terminal • Histidine kinase • Ig-like • J* • KH • KH type-2 • Kinesin motor • LIM zinc-binding 1 • LIM zinc-binding 2 • Lipoyl-binding • MPN • MTTase N-terminal • N-acetyltransferase* • NodB homology • Nudix hydrolase • Obg • PDZ • PH • PPIase FKBP-type • PPM-type phosphatase • Peptidase A1	• Peptidase M12B • Peptidase M14 • Peptidase S1* • Peptidase S8* • Protein kinase* • RNase H type-1 • Radical SAM core* • Response regulatory* • Rhodanese • Rieske • S1 motif • S1-like • SH3 • SIS • Sigma-54 factor interaction • SpoVT-AbrB 1 • SpoVT-AbrB 2 • TBDR beta-barrel • TGS • TIR • Thioredoxin • TrmE-type G • Tyr recombinase* • Tyrosine-protein phosphatase • Urease • VWFA • YjeF N-terminal • YrdC-like • bHLH • bZIP • sHSP • tr-type G

Table 8: Swiss-Prot Concepts associated with SAE features in any layer (Part A). * Indicates concept that is also associated with a neuron in any layer.

Motif		
• Beta-hairpin • DEAD box • Effector region • Histidine box-2	• JAMM motif • NPA 1 • Nudix box • PP-loop motif	• Q motif • Selectivity filter
Region		
• 3-hydroxyacyl-CoA dehydrogenase • A • Adenylyl removase • Adenylyl transferase • Basic motif • Disordered* • Domain II • FAD-dependent cmm(5)s(2)U34 oxidoreductase • Framework-3 • Interaction with substrate tRNA		• Large ATPase domain (RuvB-L) • N-acetyltransferase • NBD2 • NMP • Pyrophosphorylase • Ribokinase • Small ATPase domain (RuvB-S) • Uridyl-removing • Uridyltransferase

Table 9: Swiss-Prot Concepts associated with SAE features in any layer (Part B). * Indicates concept that is also associated with a neuron in any layer.

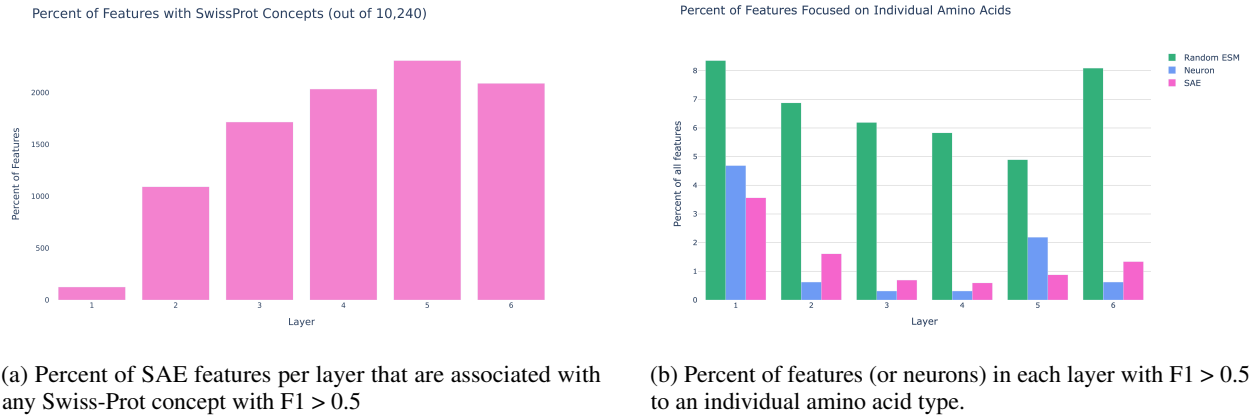


Figure 9: Additional feature-concept analysis across layers